Requirements Document

1 Project Overview

SafeAlert is a web-based Disaster Management System designed to provide real-time, location-based disaster alerts to citizens and administrative authorities across India. The system ensures timely communication, secure access, and reliable alert dissemination using email notifications.

2 Objectives

- Provide real-time disaster alerts to users based on location
- Enable administrators to create, verify, schedule, and dispatch alerts
- Ensure secure authentication and role-based access
- Maintain alert history and user records

3 Stakeholders

- End Users (Citizens)
- System Administrators
- Government / Disaster Management Authorities
- Developers & Testers

4 Functional Requirements

User Module

- User registration with email and password
- Secure login and logout
- Location selection (50 Indian cities + manual entry)
- View disaster alerts for selected location
- Filter alerts by severity and status
- Receive email notifications
- Password reset using OTP (6-digit, 10-minute validity)

Admin Module

- Admin registration using referral code
- Admin login/logout
- Create disaster alerts (type, location, severity, description)
- Schedule future alerts
- Verify alerts (two-stage verification)
- Dispatch alerts via bulk email
- View user list and search users
- View alert statistics (pending, verified, sent)

System Requirements

- Role-based access control (User/Admin)
- RESTful API support
- Logging of system actions
- SQLite database persistence

5 Non-Functional Requirements

- Security: Password hashing, session management, OTP verification
- Performance: Handle multiple users and bulk emails efficiently
- Usability: Responsive UI using Bootstrap 5
- Reliability: ACID-compliant database
- Portability: Runs on any system with Python 3.8+

6 Technology Stack

- Backend: Flask (Python)
- Frontend: HTML, CSS, Bootstrap 5, JavaScript
- Database: SQLite
- Email: Gmail SMTP
- Hosting: PythonAnywhere (optional)